Adrian Simon Bercovici

Senior Software Architect | High-Performance Distributed Systems

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LinkedIn | GitHub | StackOverflow

Summary

Software Architect with 9+ years of experience building actor-based and event-driven architectures, workflow orchestration engines, high-throughput backend services, messaging and streaming pipelines • DDD • Functional Programming and distributed systems

Professional Experience

🔷 **Senior Software Architect — Global IP Risk Scoring Platform** (Octomber 2025 – Present)

Developed a high-performance **IP Intelligence and Scoring API** designed for real-time fraud detection and risk assessment.
Engineered a multi-tier caching strategy and asynchronous background workers to handle **4,000 req/s** with ultra-low latency (**25ms p99**), featuring behavioral and anomaly detection.
Built a comprehensive observability suite with real-time dashboards for throughput, cache efficiency, and per-user rate limiting.
Rust, Actix-Web, Tokio, Redis, Clickhouse, Docker, k6

🔷 **Rust Developer (Contract)** (Mar 2025 – Octomber 2025)

Building a real-time DAW SaaS platform for audio mixing, streaming, and track management using Rust and an actor-based architecture.
Designed backend with **Kameo** (actors), **Actix-Web**, and **Tokio** for async audio control (play, pause, record, seek) and user-specific track operations via REST and WebSockets.
Rust, Kameo (actor model), Actix-Web, Tokio, async/await, WebSocket, REST

🔷 **Lead Platform Developer | Absa Bank (Contract)** (Mar 2024 – Mar 2025)

Led a team of 6 Python developers in building **Temporal.io workflows** for orchestrating long-running infrastructure operations (e.g. VM restarts, DB upgrades).
Drove a major **platform refactor** to align with **DDD** and **Dependency Injection**—owning Jira scoping, PR reviews, and pair programming sessions.
Tech: Python • Temporal.io • .NET/C# • MassTransit • PostgreSQL • CockroachDB

🔷 **.NET Software Architect | Nuvei (Fintech)** (May 2021 – Mar 2024)

Re-Architectured **high-performance payment processing engine** using Erlang/OTP and **Microsoft Orleans**, achieving **25x throughput improvement**
Redesigned **low-latency sanctions screening system** from monolith to event-driven microservices architecture.
Tech: Erlang/OTP • C#/.NET • Microsoft Orleans • Kafka • Azure • Redis • Event Sourcing

🔷 **Senior .NET Developer | TransPerfect** (Oct 2019 – May 2021)

Built **high-concurrency collaborative platform** with synchronized multi-user document editing.

Technical Skills

💻 **Stack(s)**: Rust • Erlang • .NET • Python
🧠 **Orchestration & Actors** : Temporal.io • Microsoft Orleans • Kameo (Rust) • Erlang/OTP
📡 **Real-time**: WebSockets • WebRTC • SignalR • Pub/Sub
⚡ **Messaging**: Kafka • RabbitMQ • Redis
🗄️ **Data**: PostgreSQL • MS SQL • MongoDB
☁️ **Cloud & DevOps & Obs**: Azure • Docker • Kubernetes • Grafana • Prometheus

Key Projects & Open Source

🚀 **Audio DSP (Rust)**
High-performance real-time audio processing pipeline

🗨️ **Adi-Chat**
Scalable Erlang/OTP chat server with WebSocket clustering

💬 **Ctesiphon**
.NET event-driven chat platform using Redis pub/sub

📡 **Oituz Framework**
Native Erlang WebRTC implementation

🏦 **Banking System (Erlang)**
Concurrent banking operations with actor-based account management

📄 **IEEE Research Publication**
"Real-time Architecture for Collaborative Applications"

Education

🎓 **MSc Real-time Control Systems**, Univ. POLITEHNICA Bucharest
🎓 **BSc Automatic Control & Systems Engineering**,

Implemented **WebSocket clustering** and **conflict resolution algorithms** for simultaneous user interactions.
Tech: C# • WebSockets • Redis Pub/Sub • RabbitMQ • MSSQL • ABBYY OCR

- **.NET Developer | Orange** (Feb 2018 – Oct 2019)
  Developed **distributed system management platform** for enterprise infrastructure control.
  Tech: .NET Core • Blazor • Redis • Angular
- **Software Engineer | INDAS Tech** (Jun 2017 – Dec 2017)
  Built **industrial monitoring systems** for railway infrastructure with PLC integration and sensor data processing.
  Tech: C# • WinForms • OPC DA • SCADA/MES • Allen Bradley PLCs

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